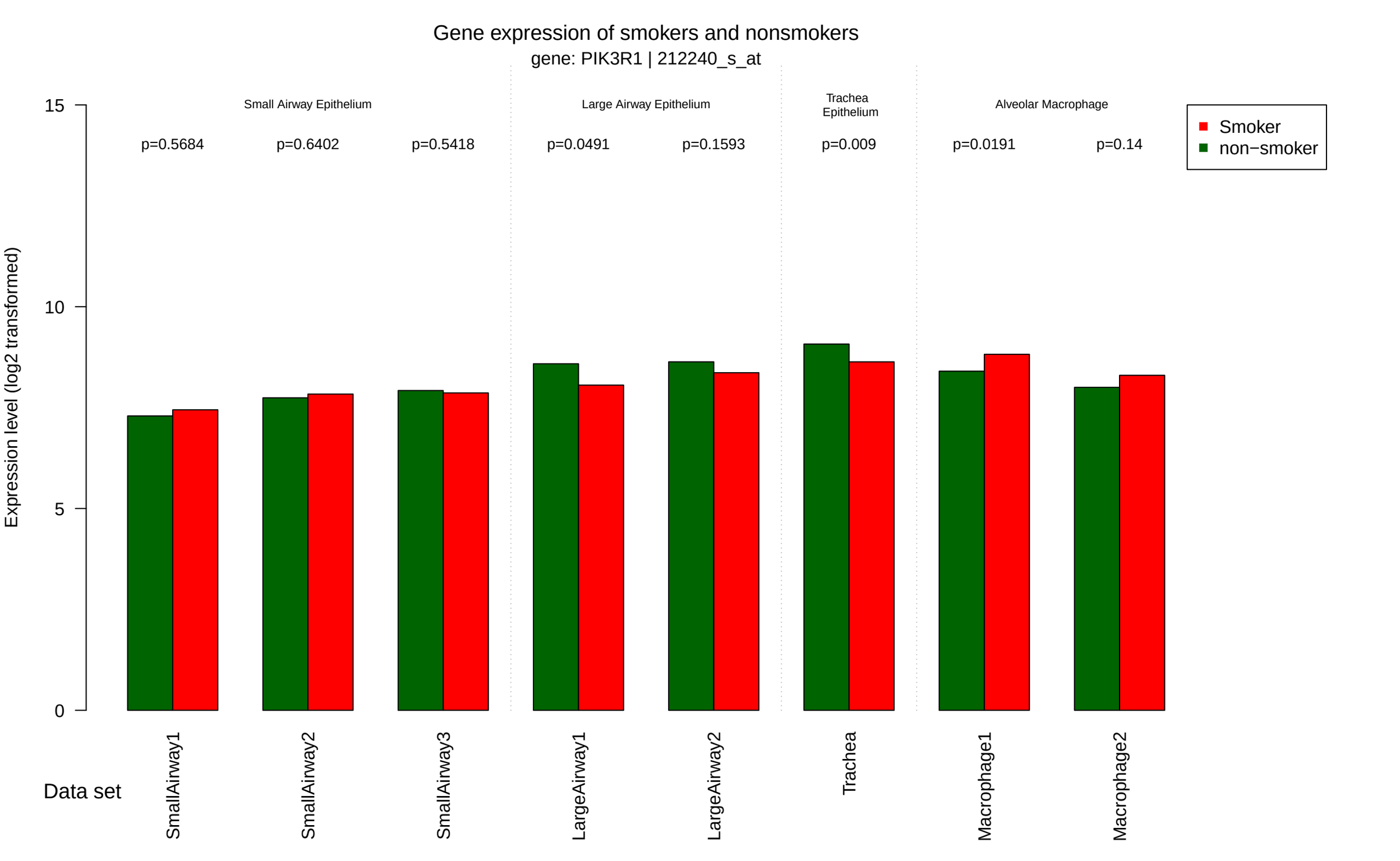
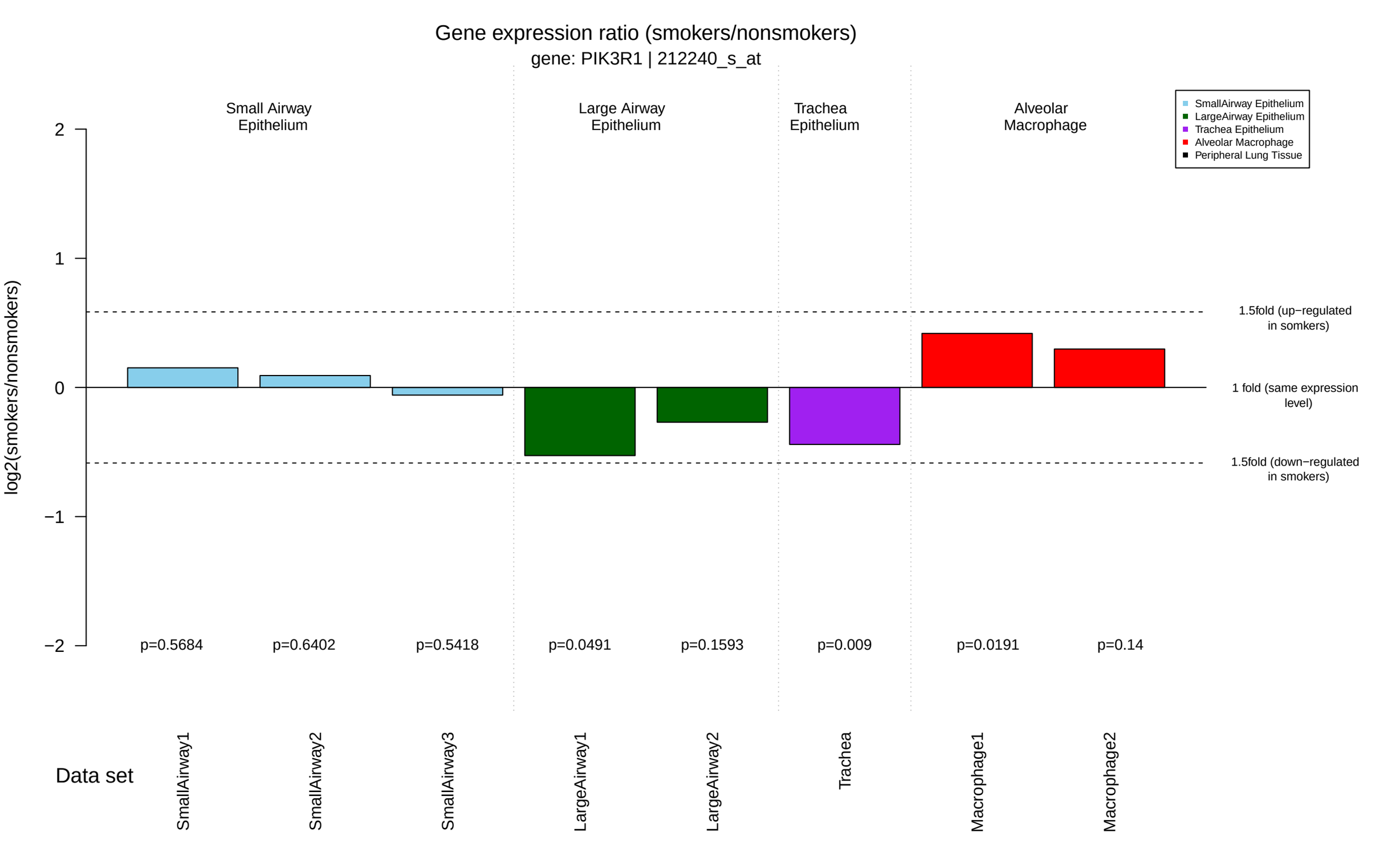
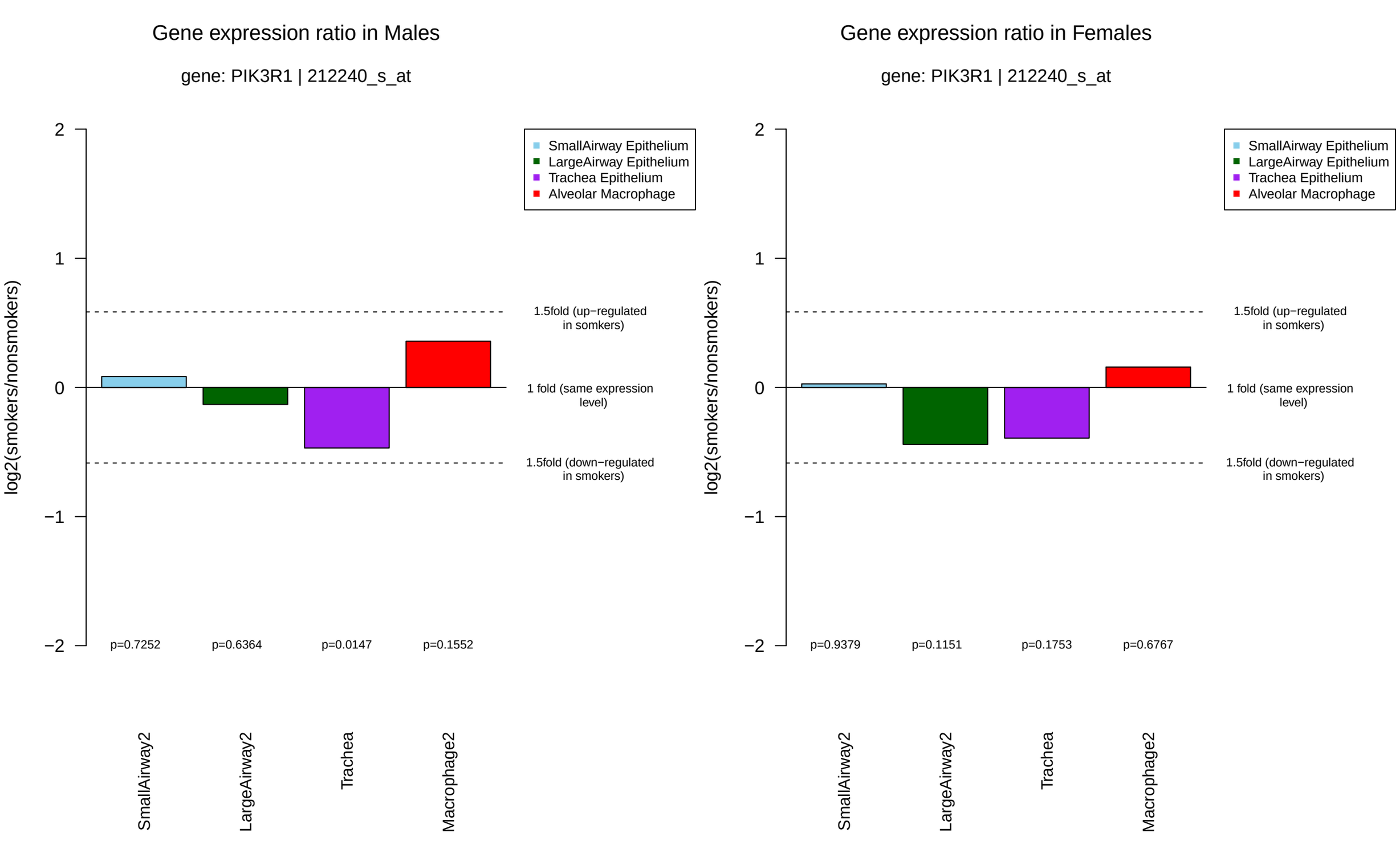
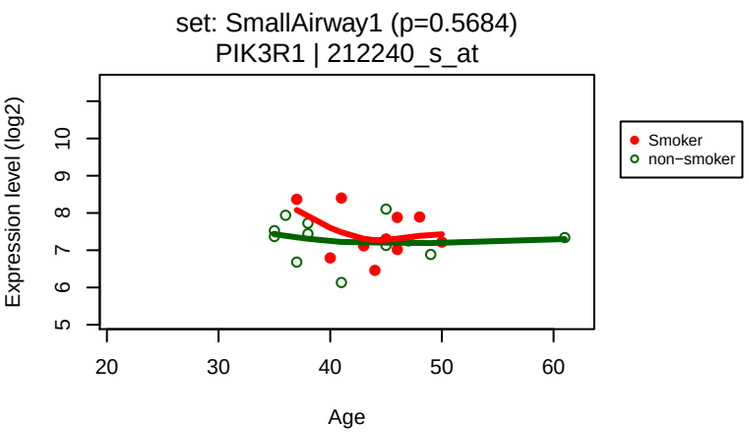
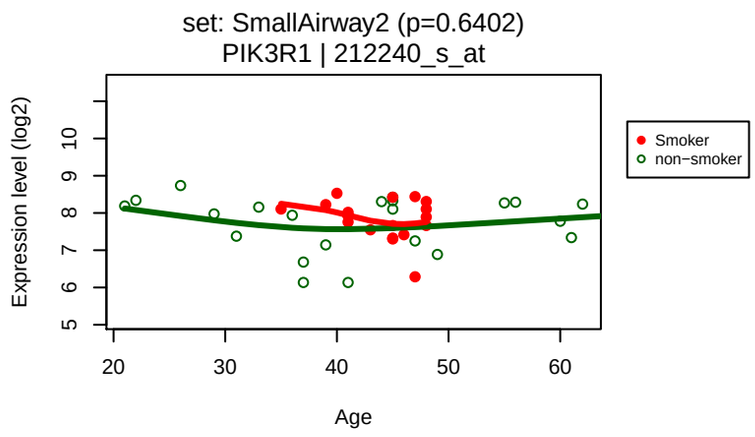
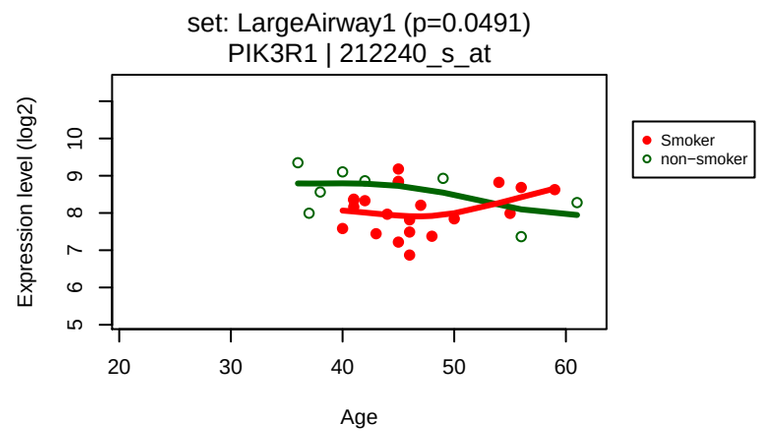

Gene expression of smokers and nonsmokers
gene: PIK3R1 | 212240_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5684
p=0.6402
p=0.5418
p=0.0491
p=0.1593
p=0.009
p=0.0191
p=0.14
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

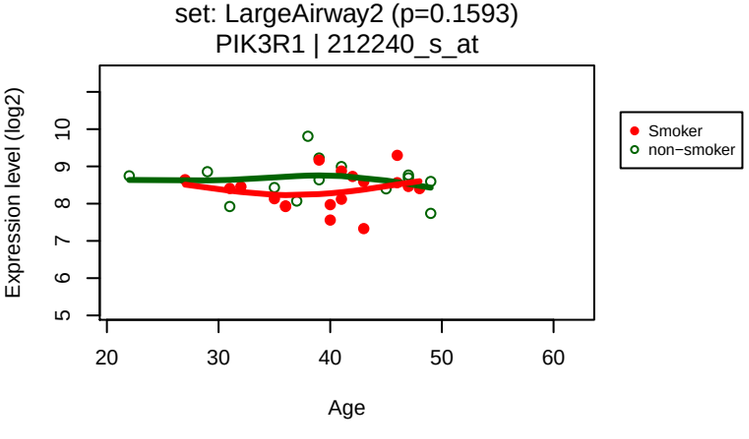

Gene expression ratio (smokers/nonsmokers)
gene: PIK3R1 | 212240_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5684
p=0.6402
p=0.5418
p=0.0491
p=0.1593
p=0.009
p=0.0191
p=0.14
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

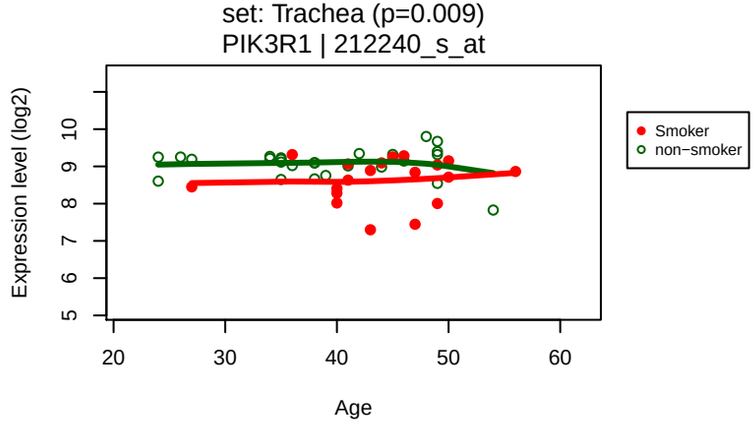

Gene expression ratio in Males
gene: PIK3R1 | 212240_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7252
p=0.6364
p=0.0147
p=0.1552
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PIK3R1 | 212240_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9379
p=0.1151
p=0.1753
p=0.6767
SmallAirway2
LargeAirway2
Trachea
Macrophage2

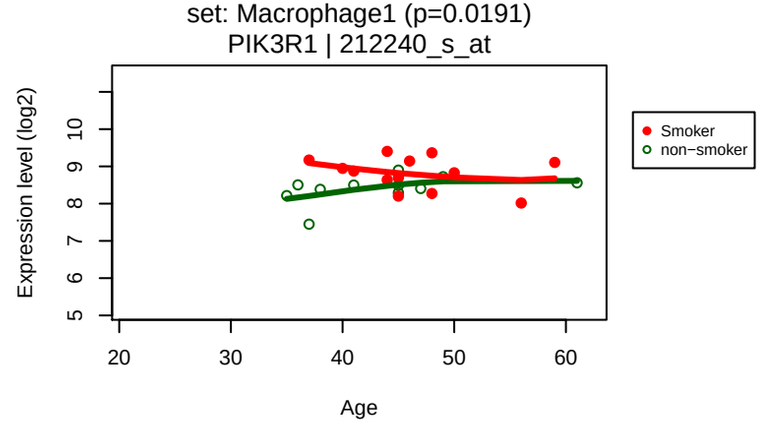

set: SmallAirway1 (p=0.5684)
PIK3R1 | 212240_s_at
Expression level (log2)
Age
Smoker
non−smoker

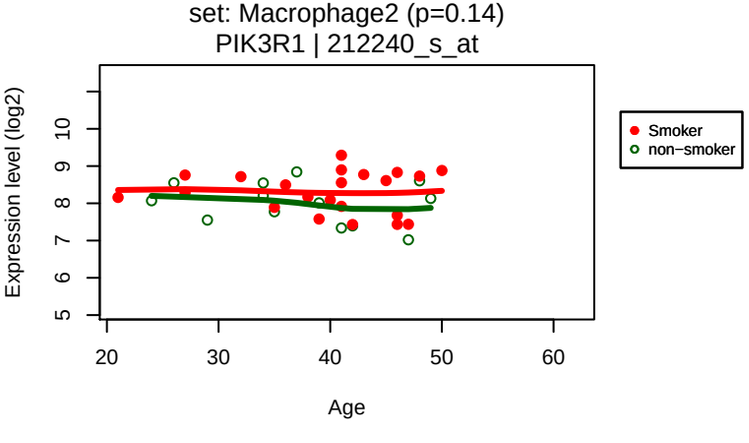

set: SmallAirway2 (p=0.6402)
PIK3R1 | 212240_s_at
Expression level (log2)
Age
Smoker
non−smoker

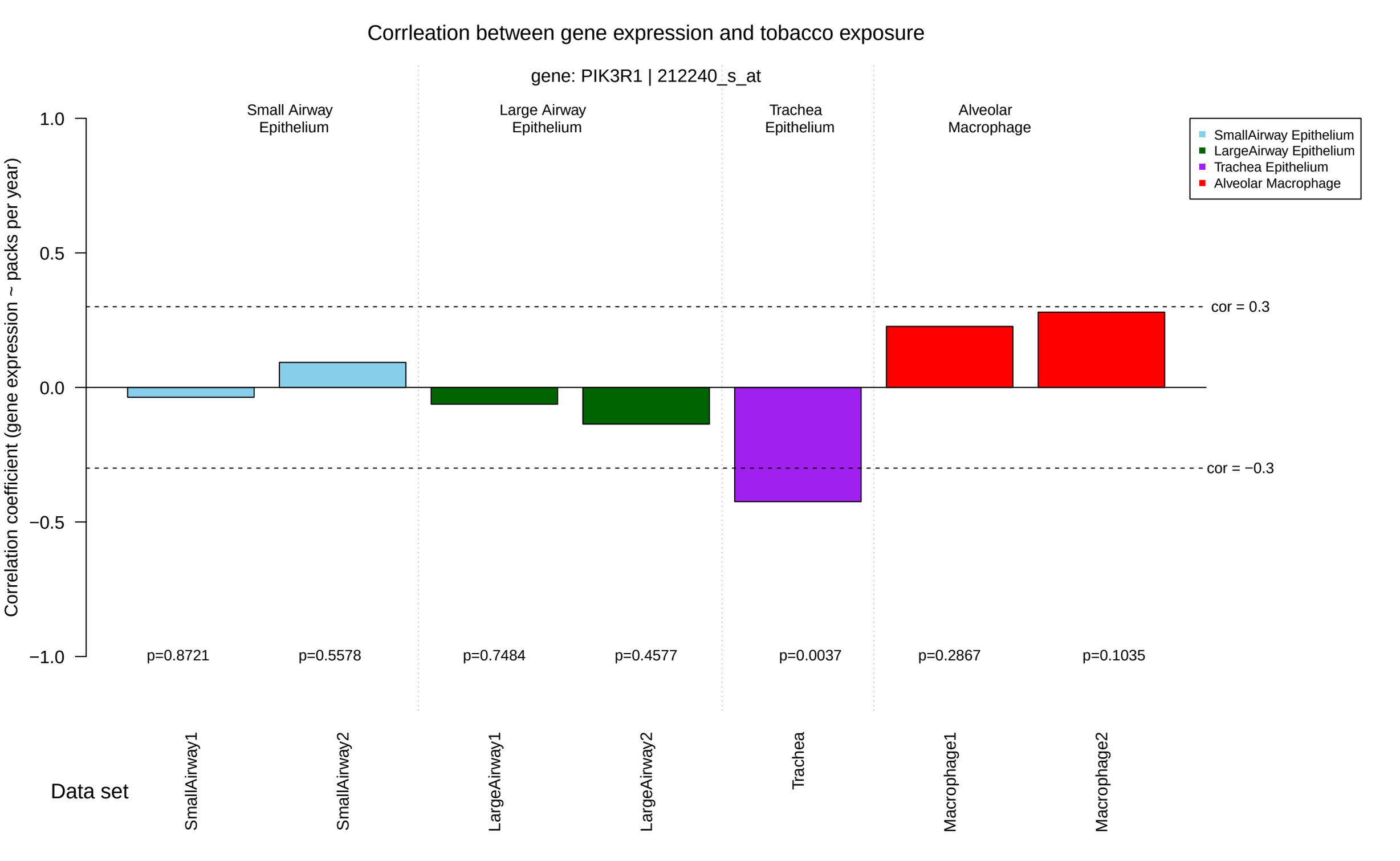

set: LargeAirway1 (p=0.0491)
PIK3R1 | 212240_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1593)
PIK3R1 | 212240_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.009)
PIK3R1 | 212240_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0191)
PIK3R1 | 212240_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.14)
PIK3R1 | 212240_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PIK3R1 | 212240_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8721
p=0.5578
p=0.7484
p=0.4577
p=0.0037
p=0.2867
p=0.1035
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2